Advanced Mathematical Techniques
Problem Set 3

1. The cosine function has infinite product representation:

$$\cos x = \prod_{k=0}^{\infty} \left[1 - \frac{4x^2}{(2k+1)^2\pi^2}\right]$$

   (a) Explain why this product converges for all x except those for which it is 0.

   This happens because, for large k, the product behaves like $-\frac{1}{k^2}$. For any infinite product, we know that:

$$\prod_{k=0}^{\infty}(1+a_k) \text{ converges iff } \sum_{k=0}^{\infty} a_k \text{ converges.}$$

   Thus, since the infinite product behavior described above is the p-series with $p = 2$, it converges, which means that the product also converges. However, this cannot be true when the product itself is 0, because then it's a divergent product to 0. Therefore, the product is only 0 when:

$$x = \frac{(2k+1)\pi}{2}$$

   At all other values of x, the series converges.

   (b) Take the logarithm of this product expansion and expand each of the logarithms to get a Maclaurin series for $\ln \cos x$. Express the coefficients of this expansion in terms of a Dirichlet series.

$$\begin{aligned}
\ln \cos x &= \sum_{k=0}^{\infty} \ln\left(1 - \frac{4x^2}{(2k+1)^2\pi^2}\right) \\
&= -\sum_{k=0}^{\infty}\sum_{j=1}^{\infty} \frac{1}{j}\left(1 - \frac{4x^2}{(2k+1)^2\pi^2}\right)^j \\
&= -\sum_{n=1}^{\infty} \frac{(4x^2)^n}{n\pi^{2n}} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^{2n}} \\
&= \boxed{-\sum_{n=1}^{\infty} \frac{(4x^2)^n}{n\pi^{2n}}(1-2^{-2n})\zeta(2n)}
\end{aligned}$$

   (c) Differentiate the result from part (b) to determine a Maclaurin series for another well-known trigonometric function in terms of Dirichlet series.

$$\frac{d}{dx}(\ln \cos x) = -\tan x$$

$$\boxed{\tan x = \sum_{n=1}^{\infty} \frac{2(4^n)x^{2n-1}}{\pi^{2n}}(1-2^{-2n})\zeta(2n)}$$

2. Concerning the Maclaurin series for the logarithm of the Gamma function,

$$\ln\Gamma(1+z) = -\gamma z + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} z^k$$

for $z < 1$.

(a) Substitute $z = -\frac{1}{2}$ into this equation to determine an interesting relationship between π and the values of the ζ function. Does your resulting series converge absolutely or conditionally?

Dropping the alternating element to prove absolute convergence:

$$\ln\left(\Gamma\left(\frac{1}{2}\right)\right) = \frac{\gamma}{2} + \sum_{k=2}^{\infty} \frac{\zeta(k)}{k2^k}$$

$$\ln\left(\sqrt{\pi}\right) = \frac{\gamma}{2} + \sum_{k=2}^{\infty} \frac{\zeta(k)}{k2^k}$$

Since $\zeta(k) \sim 1$ as $k \to \infty$, the series behaves like

$$\sum_{k=2}^{\infty} \frac{1}{k2^k}$$

which, as shown by the ratio test:

$$\lim_{k\to\infty} \left| \frac{1}{(k+1)2^{k+1}} \cdot \frac{k2^k}{1} \right| = \lim_{k\to\infty} \left| \frac{k}{2(k+1)} \right| = \frac{1}{2}$$

Thus, this series $\boxed{\text{converges absolutely}}$.

(b) Substitute $z = 1$ into this equation to determine an interesting relationship between γ and the values of the ζ function. Does your resulting series converge absolutely or conditionally?

$$\ln(\Gamma(2)) = 0 = -\gamma + \sum_{k=2}^{\infty} \frac{(-1)^k}{k} \zeta(k)$$

$$\gamma = \sum_{k=2}^{\infty} \frac{(-1)^k}{k} \zeta(k)$$

In the long run, by the other things we've shown above, this will approach $\frac{1}{k}$, which is the harmonic series, which needs alternation to converge. Therefore, this is $\boxed{\text{conditionally convergent}}$

3. Use the recurrence relation for the Gamma function to determine the value of $\Gamma(\frac{13}{3})$ in terms of $\Gamma(\frac{1}{3})$.

$$\Gamma\left(\frac{13}{3}\right) = \frac{10}{3} \cdot \Gamma\left(\frac{10}{3}\right) = \frac{10}{3} \cdot \frac{7}{3} \cdot \Gamma\left(\frac{4}{3}\right) = \frac{10 \cdot 7 \cdot 4 \cdot 1}{3^4} \Gamma\left(\frac{1}{3}\right) = \boxed{\frac{280}{81} \Gamma\left(\frac{1}{3}\right)}$$

4. Use the integral representation of the Gamma function to determine the value of the integral $\int_0^\infty x^2 \sqrt{x} e^{-2x} dx$. You may include various rational number powers of 2 and/or π in your result.

We derived in class that:

$$\int_0^\infty x^{m-1}e^{-ax^n}dx = \frac{1}{na^{\frac{m}{n}}}\Gamma\left(\frac{m}{n}\right)$$

Plugging in, we get:

$$\int_0^\infty x^{\frac{5}{2}}e^{-2x}dx = \frac{1}{1\cdot 2^{\frac{7}{2}}}\Gamma\left(\frac{7}{2}\right) = \frac{1}{8\sqrt{2}}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\Gamma\left(\frac{1}{2}\right) = \boxed{\frac{15\sqrt{\pi}}{64\sqrt{2}}}$$

5. Use the integral representation of the Gamma function to determine the value of the integral $\int_0^\infty xe^{-2x^3}dx$. You may include a value of the Gamma function evaluated between 0 and $\frac{1}{2}$ in your answer, along with various rational number powers of 2 and/or π.

Using the same logic as above,

$$\int_0^\infty xe^{-2x^3} = \frac{1}{3\cdot 2^{\frac{2}{3}}}\Gamma\left(\frac{2}{3}\right) = \frac{\sqrt[3]{2}}{6}\Gamma\left(\frac{2}{3}\right)$$

Further,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$$

Thus,

$$\Gamma\left(\frac{2}{3}\right) = \frac{\pi}{\sin(\pi\cdot\frac{1}{3})\Gamma\left(\frac{1}{3}\right)} = \frac{2\pi}{\sqrt{3}\Gamma\left(\frac{1}{3}\right)}$$

Therefore,

$$\boxed{\int_0^\infty xe^{-2x^3} = \frac{2\pi\sqrt[3]{2}}{6\sqrt{3}\Gamma(\frac{1}{3})}}$$

6. This problem concerns products of the form:

$$\prod_{n=1}^{\infty}\left(1\pm\frac{x}{n}\right)$$

(a) Write out the first 5 terms in the product $P_N = \prod_{n=1}^{N}\left(1+\frac{1}{n}\right)$ as improper fractions in order to find a pattern to determine a closed-form expression for P_N. What happens as N tends to ∞?

$$\text{First 5 terms: } (1+1), \left(1+\frac{1}{2}\right), \left(1+\frac{1}{3}\right), \left(1+\frac{1}{4}\right), \left(1+\frac{1}{5}\right)$$

As N tends to ∞, the terms tend to 1.

(b) Write out the first 5 terms in the product $Q_N = \prod_{n=2}^{N}\left(1-\frac{1}{n}\right)$ as fractions in order to find a pattern to determine a closed-form expression for Q_n. What happens as N tends to ∞?

$$\text{First 5 terms: } \left(1-\frac{1}{2}\right), \left(1-\frac{1}{3}\right), \left(1-\frac{1}{4}\right), \left(1-\frac{1}{5}\right), \left(1-\frac{1}{6}\right)$$

As N tends to ∞, the terms approach 1.

(c) Explain why the the product at the beginning of this problem should be understood as *divergent* for all values of x.

Regardless of the value of (presumably nonzero) x, if the sign chosen is $+$, then this becomes a product of $1 + H_n$, which diverges because H_n diverges. If the sign picked is $-$, then the first term is 0, meaning the series diverges to 0.

(d) Explain why the product $A = \prod_{n=1}^{\infty} \left[\left(1 + \frac{x}{n}\right) e^{-\frac{x}{n}}\right]$ *is* convergent for all x except negative integers even though the original product in this problem diverges.

$$\ln A = \sum_{n=1}^{\infty} \left[\ln\left(1 + \frac{x}{n}\right) - \frac{x}{n}\right] = \sum_{n=1}^{\infty} -\frac{x^2}{2n^2} + \frac{x^3}{3n^4} - \frac{x^4}{4n^4} + \cdots = \sum_{n=1}^{\infty} -\frac{x^2}{2n^2} + O\left(\frac{1}{n^3}\right)$$

Since the p-series with $p = 2$ converges absolutely, so deos the product. This is because the $e^{-\frac{x}{n}}$ is used as a dampening factor that eliminates the divergent nature of the product.

(e) Explain why the product $B = \prod_{n=1}^{\infty} \left[(1 + \frac{x}{n}) \ln\left(\frac{e}{1+\frac{x}{n}}\right)\right]$ *is* convergent for all positive numbers x even though the original product in this problem diverges. Would this product converge if the e was replaced by a 2? Explain.

$$\prod_{n=1}^{\infty} \left[(1 + \frac{x}{n}) \ln\left(\frac{e}{1 + \frac{x}{n}}\right)\right] = \prod_{n=1}^{\infty} \left[\left(1 + \frac{x}{n}\right) \left(1 - \ln\left(1 + \frac{x}{n}\right)\right)\right]$$

Expanding, we get:

$$1 - \frac{x^2}{2n^2} + O(\frac{1}{n^3})$$

Since $\sum_{n=1}^{\infty} \frac{x^2}{2n^2}$ converges by p-series, we know that the series converges for all $x > 0$.If e were replaced by 2, then the factor would approach $\ln(2) \neq 1$ as $n \to \infty$, so the infinite product would diverge.

(f) Explain why part (d) may be a better choice to provide an expansion that converges than (e). It might be good to think about the differences between where these products are defined.

Part (d) covers all complex numbers as well, while part (e) doesn't cover complex numbers. That makes it way better because it can cover more ground in terms of being able to process more numbers.